

Evolutionary Algorithms and Numerical Optimization

MODULE I: Fundamentals of Numerical Optimization and Evolutionary Computing

Numerical Optimization

- Introduction to Optimization
- Mathematical Formulation of Optimization Problems
- Objective Functions and Constraints
- Optimization versus Constraint Satisfaction Problems
- Linear and Non-linear Optimization
- Local and Global Optimum
- Convex and Non-convex Optimization
- Numerical Optimization Techniques
- Gradient-Based and Derivative-Free Optimization Methods

Introduction to Evolutionary Computing

- Evolutionary Computing Metaphor
- Optimization, Modelling and Simulation Problems
- Famous NP Problems
- Example Applications of Evolutionary Algorithms
- Components of Evolutionary Algorithms
- Evolutionary Search Process
- Fitness Evaluation
- Population Initialization
- Representation Techniques
 - Binary Representation
 - Integer Representation
 - Real-Valued Representation
 - Permutation Representation
 - Tree Representation
- Selection, Parent Selection, Recombination and Mutation Operators
- Survivor Selection
- Constraint Handling Techniques

MODULE II: Evolutionary Algorithms and Swarm Intelligence

Evolutionary Algorithms

- Genetic Algorithms
- Evolution Strategies
- Evolutionary Programming
- Genetic Programming
- Learning Classifier Systems
- Differential Evolution
- Estimation of Distribution Algorithms
- Parameter Control in Evolutionary Algorithms

Swarm Intelligence

- Particle Swarm Optimization
- Search Behaviour and Convergence
- Performance Analysis
- Benchmark Optimization Problems
- Comparative Analysis of Evolutionary Algorithms

Hybrid Evolutionary Algorithms

- Memetic Algorithms
- Design Issues in Memetic Algorithms
- Non-Stationary and Noisy Function Optimization
- Characterisation
- Effect of Drift
- Sources of Uncertainty

MODULE III: Multi-objective Optimization and Advanced Evolutionary Techniques

Multi-objective Numerical Optimization

- Fundamentals of Multi-objective Optimization
- Pareto Optimality
- Dominance Concepts
- Evolutionary Approaches to Multi-objective Optimization
- Constraint Handling in Multi-objective Optimization

Advanced Evolutionary Computing

- Interactive Evolution
- Design versus Optimization
- Automatic Elicitation of User Preferences
- Co-evolutionary Systems
- Applications of Evolutionary Optimization in Data Science
- Feature Selection
- Hyperparameter Optimization
- Engineering Design Optimization
- Scheduling and Resource Allocation
- Current Trends in Evolutionary Optimization